

The WRA system was developed after a review of Australia's biosecurity systems which identified the need to screen plants for weed potential before importation.

Nairn Quarantine Review

Before 1996, plants proposed for importation into Australia were checked against a 'prohibited list' contained in a proclamation of the Quarantine Act 1908.

The 'prohibited list' included plant species that had been identified as weed threats to Australia due to their weed status elsewhere in the world and/or had known quarantine pest and disease risks associated with their importation.

Quarantine in Australia was reviewed in 1996, following a number of incursions of exotic pests and diseases (Nairn et al. 1996). The Nairn Review Committee evaluated Australia's quarantine policies and programs. The review aimed to help protect Australia's favourable pest and disease status while ensuring Australia complied with its international trade obligations.

The Committee felt there was a need to strengthen the assessment process for importing plants, particularly ornamental plants used in the nursery trade. It noted that the 'prohibited list' did not adequately reduce the introduction of new weeds. The review committee recommended "...that the regulations governing the import of seeds and plant germplasm be based on a permitted list for entry rather than solely the current prohibited list."

Development

The WRA is a question-based scoring system that determines the weed potential of plants proposed for importation. The questions address traits of the plant which are considered indicative of weed potential, as well as non-invasive characteristics. The WRA system can therefore be used to predict useful non-problematic plants as well as predicting potential weeds of the environment and/or agriculture.

The WRA system underwent calibration and testing by Australian and international weed experts before it was released for wider consultation. The system was tested with 370 species, ranging from environmental and agricultural weeds to benign and beneficial plants. It was judged on its ability to correctly reject weeds, accept non-weeds and generate a low proportion of species requiring further evaluation.

Implementation

Prior to implementation, major interest groups were consulted, including state and territory departments of agriculture, the Department of the Environment, the Weeds Cooperative Research Centre, CSIRO, the World Wildlife Fund, seed companies, the grains industry, horticultural peak bodies and weed experts. Relevant comments from the consultation phase were incorporated into the system and it was well supported by the groups.

The system has been endorsed by the Department of the Environment and meets the requirements of the Environmental Protection and Biodiversity Conservation Act 1999.

In April 1997, the system was endorsed to assess all new plants before entering Australia. Shortly thereafter, under revised legislation (the Quarantine Proclamation 1998), all plant species were prohibited from entering Australia unless they were on the Permitted Seeds List. To be added to the Permitted Seeds List, new species had to be formally assessed as having a low potential to become weeds in Australia using the WRA system.

Domestic and international evaluations

The Primary Industries Standing Committee reviewed the WRA system after seven years. The review found that the system was generally considered robust and effective.

The Australian WRA system is internationally recognised as one of the best systems to determine the potential of plant species to become weeds of agriculture and/or the environment. Numerous peer-reviewed papers have supported the implementation of the WRA system and recommended its wider application.

Keller et al. (2007) demonstrated that the WRA system produced considerable net bio-economic benefits to Australia within a decade of its implementation. Gordon et al. (2008a) researched various applications of the WRA system and found that the system rejects an average of 90 per cent of known invasive species and accepts an average of 70 per cent known non-invasive species.

Several Australian states and territories have used the WRA system as the basis for determining the weed potential of plants proposed for importation into their respective jurisdictions. The WRA system was also adopted by New Zealand (with little change) at approximately the same time as it was adopted by Australia.

Modified versions of the WRA system have been tested for use in the Bonin Islands of Japan (Kato et al. 2006; Nishida et al. 2009), Canada (McClay et al. 2010), the Czech Republic (Křivánek and Pyšek 2006), Florida (Gordon et al. 2008b), Hawai'i (Daehler and Carino 2000), Italy (Crosti et al. 2009), in the Mediterranean (Gasso et al. 2010), the Pacific Islands (Daehler et al. 2004), Tanzania (Dawson et al. 2009) and the United States (Gordon and Gantz 2008).

The Permitted Seeds List Review

The Permitted Seeds List was also created in response to the Nairn review's recommendations.

The [department finalised](#) a review of plant seeds permitted entry into Australia (Schedule 5 of the Quarantine Proclamation 1998, also known as the Permitted Seeds List) in December 2006. The review involved replacing 2,913 genus-level listings with species already present in Australia within those genera.

In conducting the review, Australia complied with its international obligations by ensuring that species already present in Australia and not under ‘official control’ remained on the Permitted Seeds List.

The review of the Permitted Seeds List was undertaken to enhance Australia’s favourable pest and disease status by preventing the importation of plant species with a high potential of becoming weeds in Australia.

Throughout the review, the department consulted extensively with stakeholders including state and Commonwealth government departments, industry representative bodies, conservation organisations, research and development corporations, co-operative research centres, retail nurseries, gardening enthusiasts and other interested stakeholders.

Permitted Seeds List under the Biosecurity Act 2015

On 16 June 2016, the Biosecurity Act 2015 replaced the Quarantine Act 1908 as the primary biosecurity legislation in Australia and the Biosecurity (Prohibited and Conditionally Non-prohibited Goods) Determination 2016 (the Goods Determination) replaced the Quarantine Proclamation 1998. The Goods Determination specifies goods or classes of goods that are prohibited or conditionally non-prohibited for the purpose of the Biosecurity Act. Under the Goods Determination, Australia continues to manage the entry of seed species via a permitted species list that is published on the department’s website.

Although the legislation underpinning the permitted seeds list has changed, the Weed Risk Assessment process continues unchanged under the new legislation.

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Footnote

‘Official control’ is defined by the International Plant Protection Convention as ‘the active enforcement of mandatory phytosanitary regulations and the application of mandatory phytosanitary procedures with the objective of eradication or containment of quarantine pests or for the management of regulated non-quarantine pests’ (FAO 2016).

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Source: <https://www.agriculture.gov.au/biosecurity-trade/policy/risk-analysis/plant/weeds/development>